

Shifting the Parent Functions

Name _____ · Aug 24

Formula bank Outside (up/down): $f(x) + k$ up k · $f(x) - k$ down k — as written
 Inside (left/right): $f(x - h)$ RIGHT h · $f(x + h)$ LEFT h — opposite **Domain & range:** slide WITH the graph

VOCABULARY

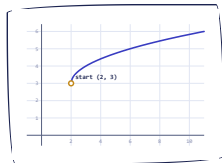
Transformation — any change made to a _____'s graph

Translation — a _____ — the shape and size stay exactly the same

EXAMPLE 1 · TYPE 1 — DESCRIBE THE SHIFT

Describe every shift of $y = \sqrt{x-2} + 3$ and name the new start point.

1. Inside: $x - 2$ moves the graph _____ (inside is opposite).
2. Outside: $+3$ moves it _____ (outside is as written).
3. The start point $(0, 0)$ lands at _____.



EXAMPLE 2 · TYPE 2 — WRITE THE EQUATION

Write the equation: $y = |x|$ shifted left 4 and down 1.

1. Left 4 goes INSIDE with the opposite sign: $|x$ _____ |.
2. Down 1 goes OUTSIDE as written: _____.
3. Equation: _____

EXAMPLE 3 · TYPE 3 — DOMAIN & RANGE AFTER THE SHIFT

State the domain and range of $y = \sqrt{x-2} + 3$.

1. The graph starts at $(2, 3)$ and runs right and up, so the domain is $x \geq$ _____.
2. The lowest output is the start height: range $y \geq$ _____.

YOU TRY

1.

Describe every shift of $y = x^3 + 5$.

2.

Write the equation: $y = \sqrt{x}$ shifted right 6 and down 2, then give its domain and range.

SUMMARY

Inside the function moves _____; outside moves _____. The domain and range _____.

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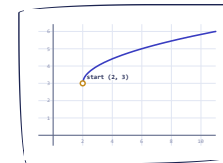
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